

Project

By: Trace Buckley, Sammy Nelms, and Jefferson Uzzle

The dataset we chose

- This dataset is from the 1985 Ward's Automotive Yearbook, which is a yearly collection of car industry data. It is collected by WARD'S AUTO
- <https://archive.ics.uci.edu/dataset/10/automobile>
- This dataset contains 26 different attributes for each car, including miles per gallon, weight, size, car type, and make. This is extremely helpful in finding correlations from each set with one another.

Our Client!

- Our persona is searching for an affordable vehicle that achieves excellent mileage on both highways and cities
- Buying a new vehicle to accommodate for his family-centered lifestyle

Frugal Fred



AGE	24
EDUCATION	B.S. in M.Eng, M.A. Psychology
STATUS	Engaged
OCCUPATION	Design engineer
LOCATION	Manassas, VA

"I need to find a new vehicle to support my grandmother, fiancé, and eventual kids with"

Personality

Introvert Thinker

Fred is a Design Engineer for a government contractor. He has his undergraduate degree in engineering and a master's in psychology. He's in a 5-year relationship from undergraduate, just proposed, and wants to settle down and have kids sometime soon. He's currently in the market for a new car. He's been driving a '72 Honda Civic with 300,000 miles on it. He does not know much about cars or how they work

Core needs

- Excellent Gas MPG
- Safe, easy to drive, and long term purchase
- Affordable to insure (moving off of parents' plan)

Frustrations

- Cars often fail after a short time nowadays
- Doesn't know what to look for in a car
- Insurance can be very expensive for new plans

Brands

Toyota Honda Hyundai
Miata Open to others

Price Range
\$10k-40k

Most Critical Factor:
Safety

Question One:

How can engine properties (stroke, compression ratio, engine type) affect mpg?

This is interesting to us as gas is currently expensive, and an engine having higher miles/gallon is an immediate benefit for that car. This question is inferential as it cannot be directly answered from the values, and multiple data sets must be looked at, such as stroke, compression ratio, engine type, and placement, to gain an understanding

Question Two:

- What/which geometric features can affect the mpg of the cars and in what ways?

This question can give insight to how the build of a car affect the mpg of said car. You might like to use this metric in determining whether particular features are worth it to you.

Question Three:

- What's the most cost-effective car?

Whether it's adding or dropping features, knowing how those features affect the buying cost of a car. Additionally, will added function of the car cause it to have high regular maintenance costs.

Question Four:

How does each car body and make affect the insurance rating and losses on each vehicle?

We are interested in this question as the data set is from an insurance company, but we are also nearing the age where we may need to insure our own vehicles and knowing what type of car has a lower insurance (and costs the insurance company less/year) will be helpful to know. This is inferential as we will be able to directly access the insurance values and scales, but must make connections ourselves between make and body.

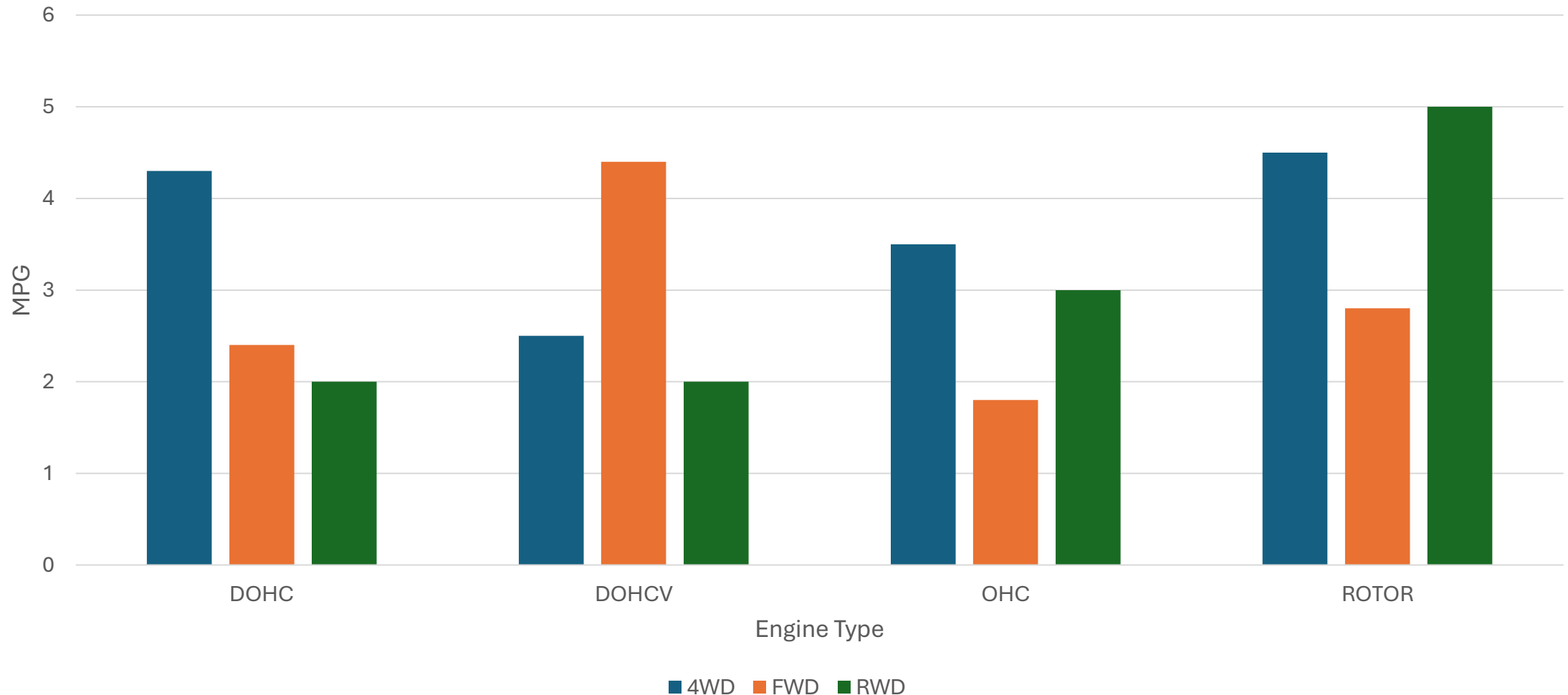
Question Five:

- What style of driving is going to have the best cost-efficiency?

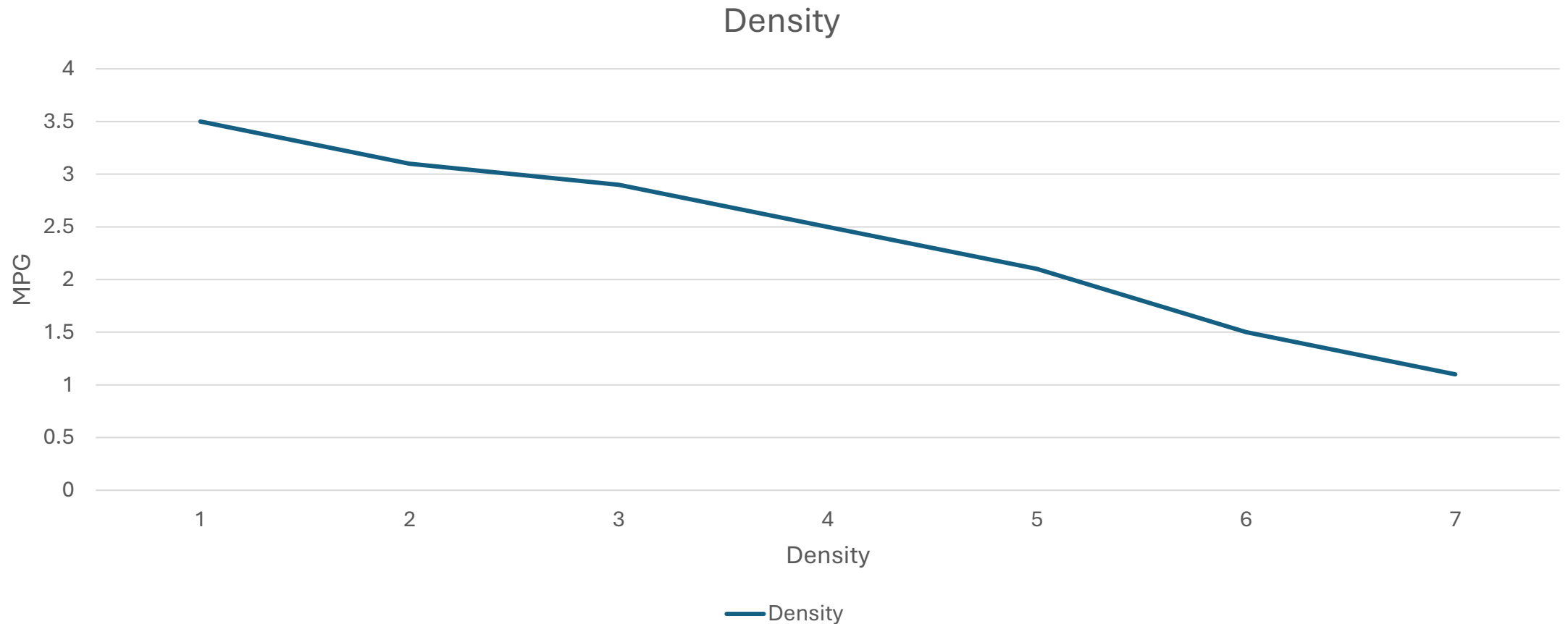
Driving in a city versus driving on the highway can have different mile per gallon ratings. Some cars may be better at city driving and some may be better at highway driving. Within the top performers in each of those categories, which ones have a more demanding price range.

Question	What aspect or feature could be used to answer this question?	<i>Does this data/feature have the potential to answer the question</i>
How can engine properties (stroke, compression ratio, engine type) affect mpg?	Engine stroke, compression ratio, engine-type, bore, fuel system, engine size, city miles per gallon, engine location, highway miles per gallon.	Yes
What/which geometric features can affect the mpg of the cars and in what ways?	Car length, width, height, curb weight, city miles per gallon, highway miles per gallon.	Yes
What's the most cost-effective car?	City miles per gallon, highway miles per gallon, horsepower, drive wheels, make, price	Yes
How does each car body and make affect the insurance rating and losses on each vehicle?	Normalized losses, make, price, body	Yes
What style of driving is going to have the best cost-efficiency?	Price, highway mile per gallon, city miles per gallon.	Yes

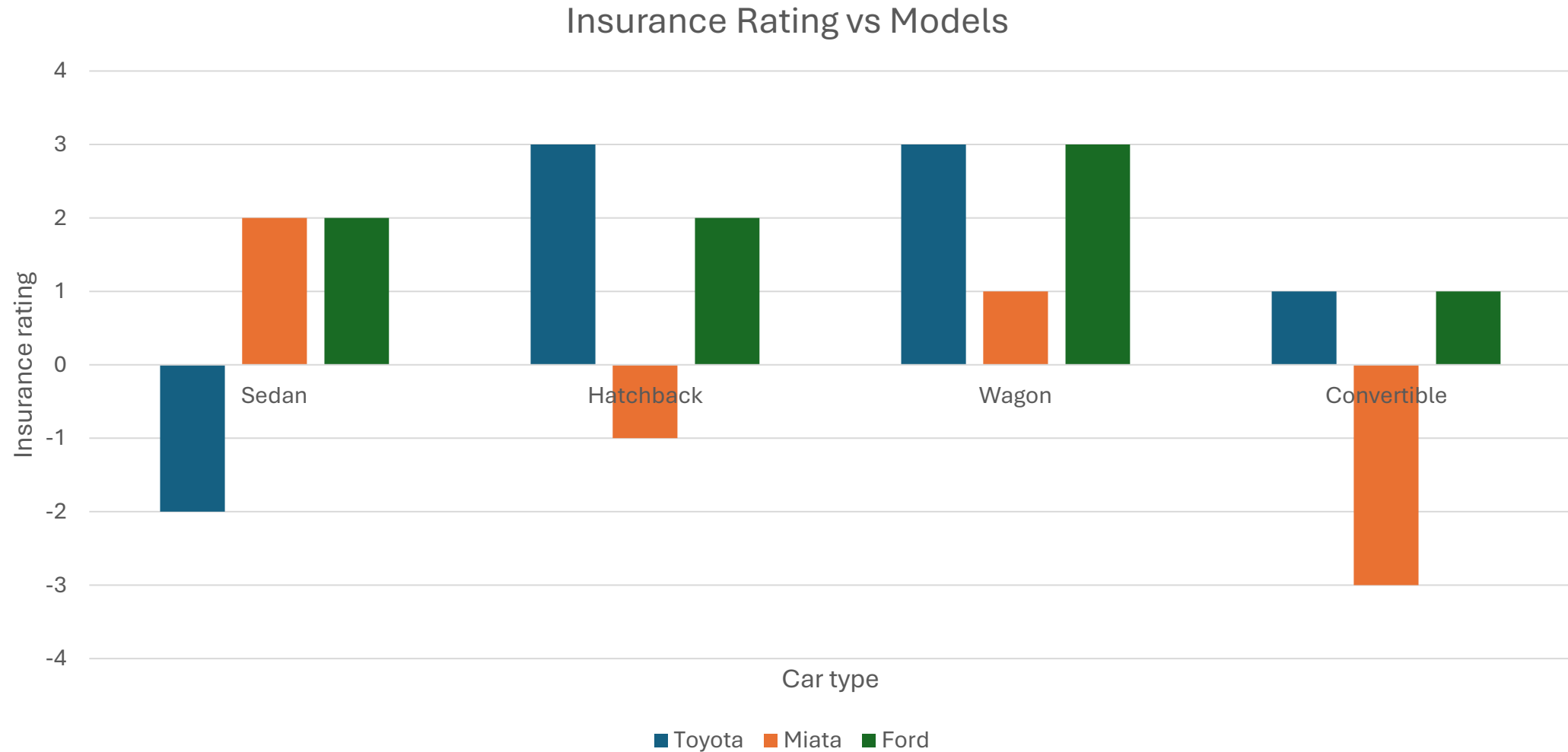
How can engine properties (stroke, compression ratio, engine type) affect mpg?



What/which geometric features can affect the mpg of the cars and in what ways?



How does each car body and make affect the insurance rating and losses on each vehicle?



How can engine properties (stroke, compression ratio, engine type) affect mpg?

- Metrics:

- Stroke
- Bore
- Engine Size
- Number of Cylinders
- Fuel Type
- Compression Ratio
- Peak RPM
- Horsepower
- Highway MPG
- City MPG

- Python Packages: Pandas, Numpy, Matplotlib
- Plots and visualization: Scatter plots, Bar Graphs

What/which geometric features can affect the mpg of the cars and in what ways?

- Metrics:
 - Car density $(\text{Length} \times \text{Width} \times \text{Height}) / (\text{Weight})$
 - Price
 - Highway and City MPG
 - Body Style
- Python Packages: Pandas, Numpy, Matplotlib
- Plots and visualization

How does each car price and make affect the insurance rating and losses on each vehicle?

- Metrics: Insurance Risk Assessment, Price, Insurance Normalized Losses
- Averages will be heavily relied upon as well as binned sorting
- Python Packages: Pandas, Numpy, Matplotlib, scipy
- Plots and visualization: Bar graphs, line plots, dual-axis plots

How can engine properties affect mpg?- Output

The question is "How can engine properties affect mpg?"

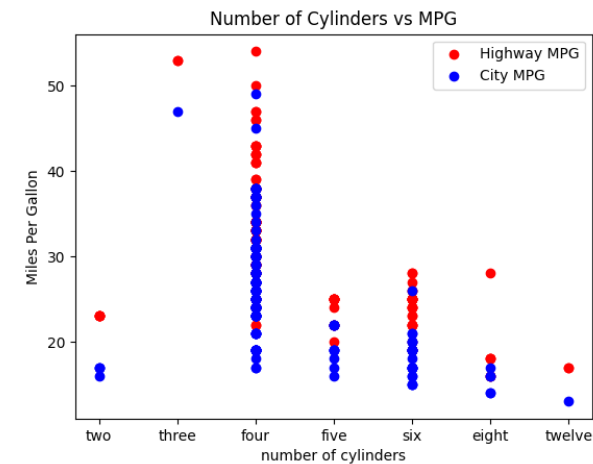
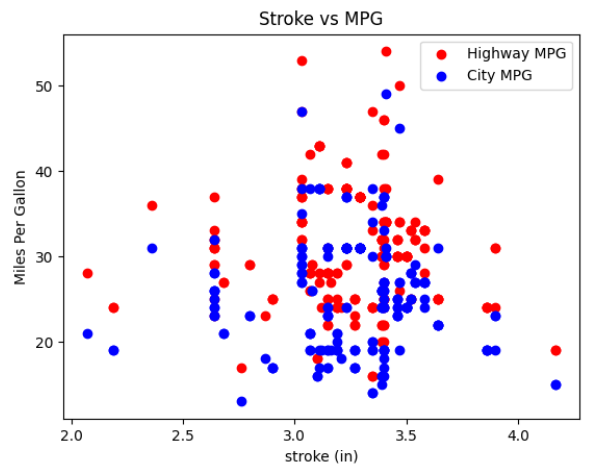
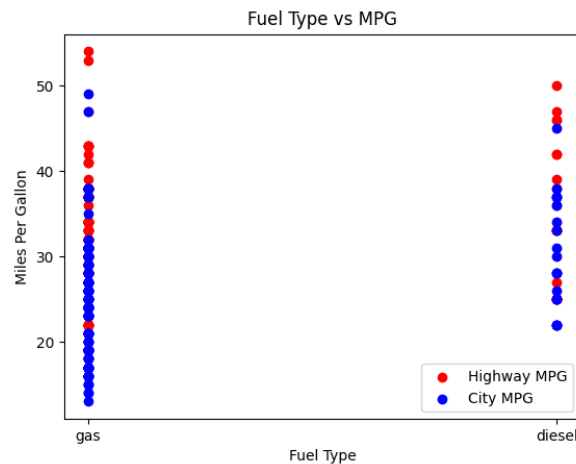
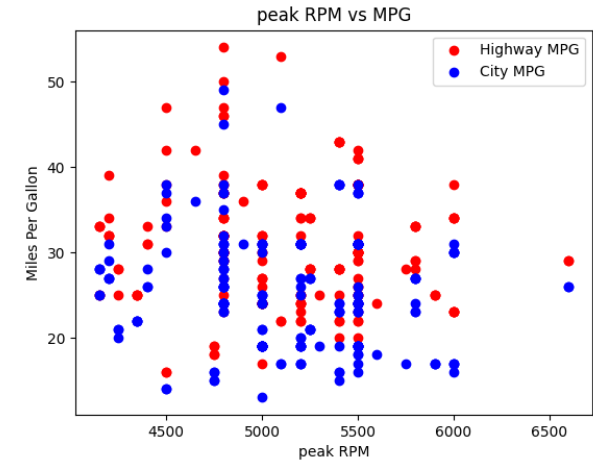
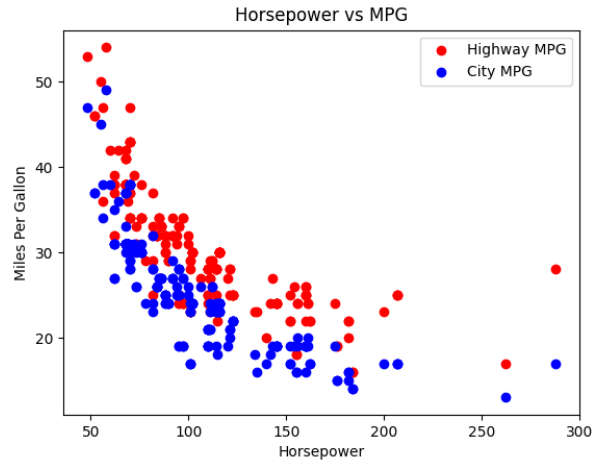
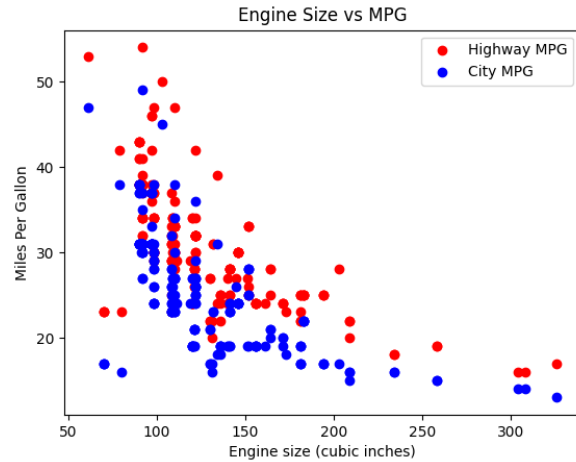
The car with the maximum MPG is a honda, which has an engine size of 92.0, a horsepower of 58.0, and four cylinders. The city MPG is 49 and the highway MPG is 54

The car with the minimum MPG is a mercedes-benz, which has an engine size of 308.0, a horsepower of 184.0, and eight cylinders. The city MPG is 14 and the highway MPG is 16

The properties which correlate to Lower MPG are larger engine size and higher horsepower

The properties which do not do a good job in predicting MPG are peak RPM, fuel type, stroke, and number of cylinders

How can engine properties affect mpg?- Plots



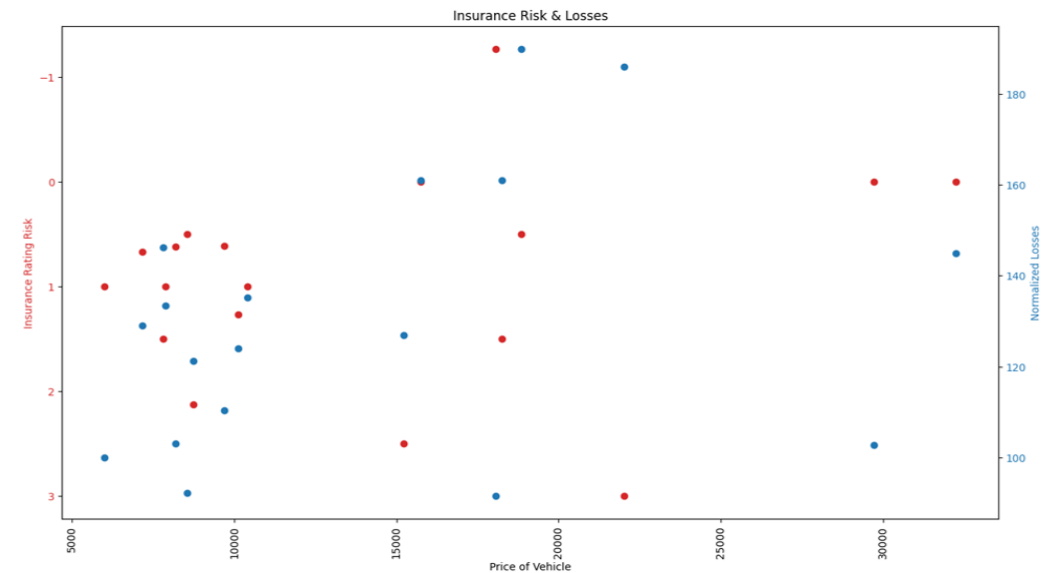
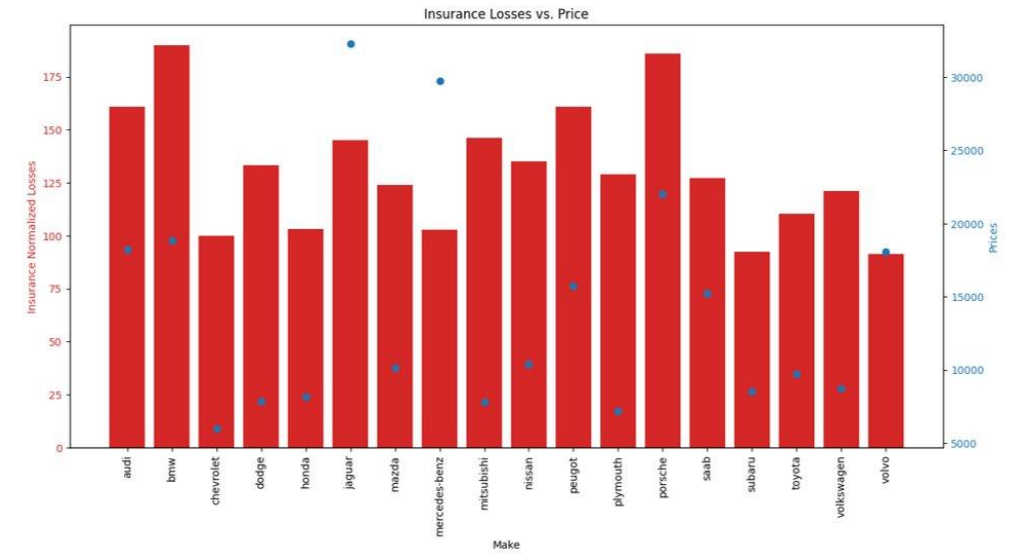
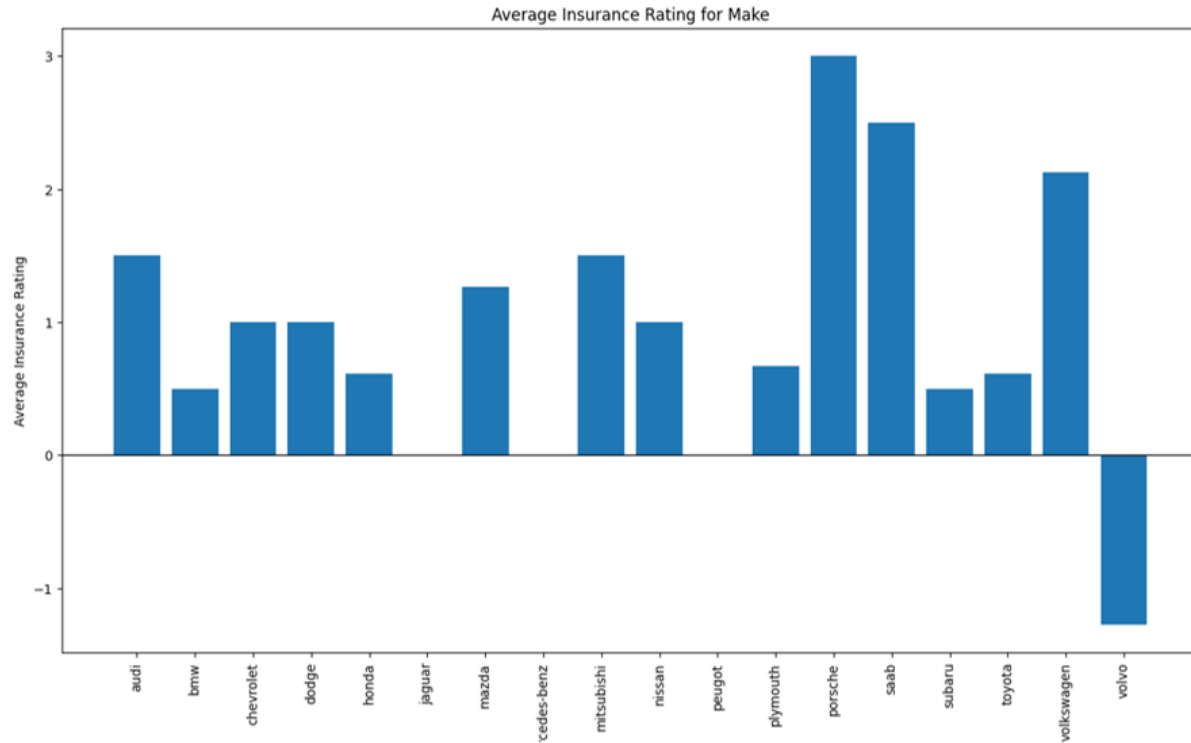
How does each car price and make affect the insurance rating and losses on each vehicle? - Output

```
audi: Insurance rating = 1.5000
audi: Insurance Losses, Normalized = 161.00
audi: Car Price = 18246.250000
bmw: Insurance rating = 0.5000
bmw: Insurance Losses, Normalized = 190.00
bmw: Car Price = 18857.500000
chevrolet: Insurance rating = 1.0000
chevrolet: Insurance Losses, Normalized = 100.00
chevrolet: Car Price = 6007.000000
dodge: Insurance rating = 1.0000
dodge: Insurance Losses, Normalized = 133.44
dodge: Car Price = 7875.444444
honda: Insurance rating = 0.6154
honda: Insurance Losses, Normalized = 103.00
honda: Car Price = 8184.692308
jaguar: Insurance rating = 0.0000
jaguar: Insurance Losses, Normalized = 145.00
jaguar: Car Price = 32250.000000
mazda: Insurance rating = 1.2667
mazda: Insurance Losses, Normalized = 123.93
```

```
mazda: Car Price = 10130.666667
mercedes-benz: Insurance rating = 0.0000
mercedes-benz: Insurance Losses, Normalized = 102.80
mercedes-benz: Car Price = 29726.400000
mitsubishi: Insurance rating = 1.5000
mitsubishi: Insurance Losses, Normalized = 146.20
mitsubishi: Car Price = 7813.000000
nissan: Insurance rating = 1.0000
nissan: Insurance Losses, Normalized = 135.17
nissan: Car Price = 10415.666667
peugeot: Insurance rating = 0.0000
peugeot: Insurance Losses, Normalized = 161.00
peugeot: Car Price = 15758.571429
plymouth: Insurance rating = 0.6667
plymouth: Insurance Losses, Normalized = 129.00
plymouth: Car Price = 7163.333333
porsche: Insurance rating = 3.0000
porsche: Insurance Losses, Normalized = 186.00
porsche: Car Price = 22018.000000
```

```
porsche: Insurance rating = 3.0000
porsche: Insurance Losses, Normalized = 186.00
porsche: Car Price = 22018.000000
saab: Insurance rating = 2.5000
saab: Insurance Losses, Normalized = 127.00
saab: Car Price = 15223.333333
subaru: Insurance rating = 0.5000
subaru: Insurance Losses, Normalized = 92.25
subaru: Car Price = 8541.250000
toyota: Insurance rating = 0.6129
toyota: Insurance Losses, Normalized = 110.29
toyota: Car Price = 9696.645161
volkswagen: Insurance rating = 2.1250
volkswagen: Insurance Losses, Normalized = 121.25
volkswagen: Car Price = 8738.125000
volvo: Insurance rating = -1.2727
volvo: Insurance Losses, Normalized = 91.45
volvo: Car Price = 18063.181818
The Pearson Correlation r_value is 0.1089
The correlation is mildly significant
```

How does each car price and make affect the insurance rating and losses on each vehicle? -Plots



What/which geometric features can affect the mpg of the cars and in what ways?

The question is "what/which geometric features can affect the mpg of the cars and in what ways?"

Makes of hatchbacks: {'plymouth', 'mazda', 'porsche', 'mercury', 'nissan', 'saab', 'audi', 'alfa-romero', 'honda', 'subaru', 'mitsubishi', 'renault', 'chevrolet', 'isuzu', 'volkswagen', 'toyota', 'dodge'}

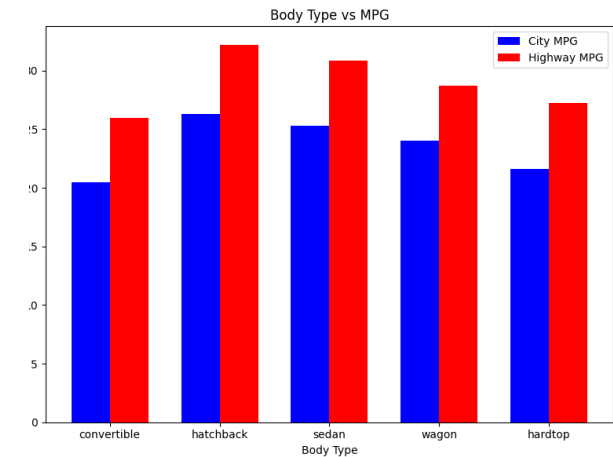
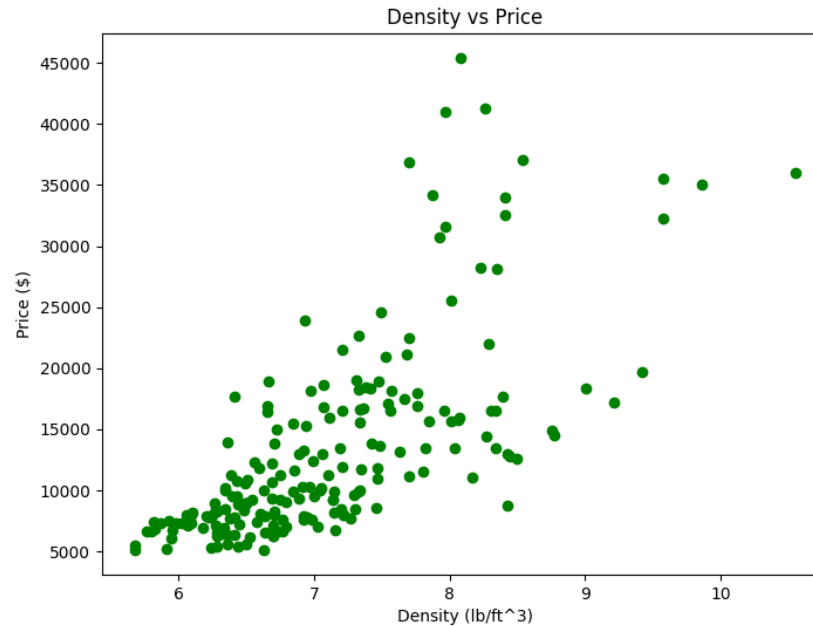
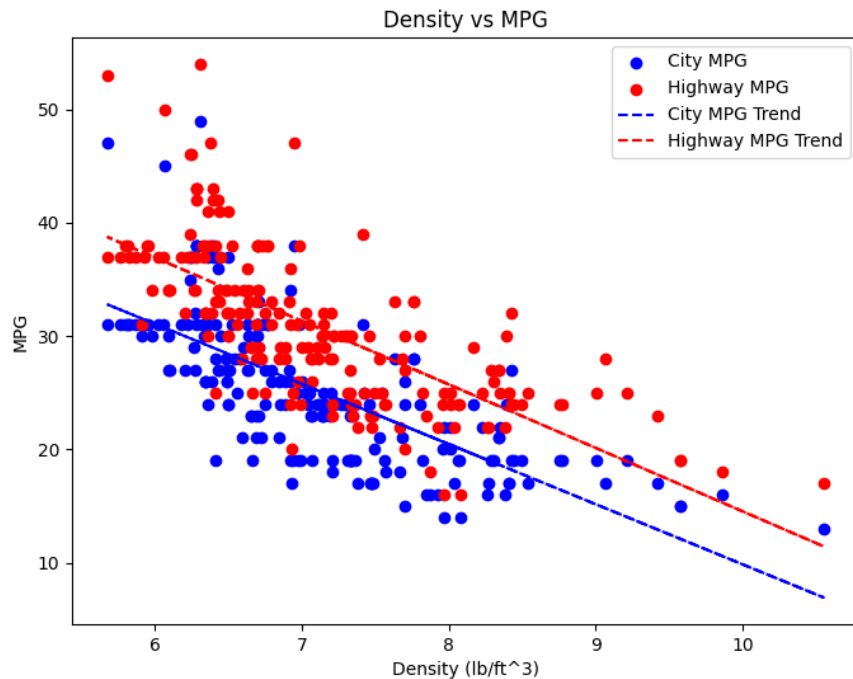
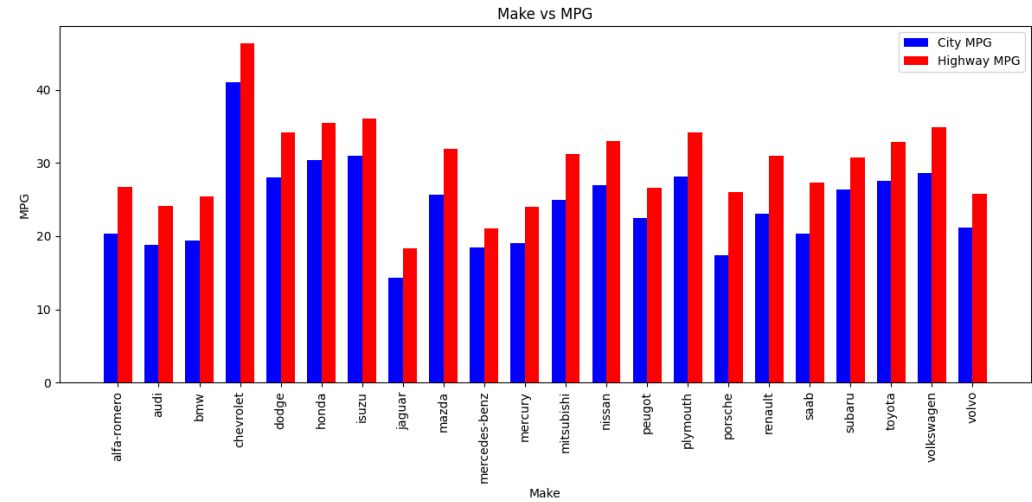
Cars that are chevrolet and hatchbacks include Car 19 and Car 20.

Additional information about these cars is as follows:

	Car 19	Car 20
symboling	2	1
normalized_losses	121	98
make	chevrolet	chevrolet
fuel_type	gas	gas
aspiration	std	std
num-of-doors	two	two
body-style	hatchback	hatchback
drive-wheels	fwd	fwd
engine_location	front	front
wheel_base	88.40	94.50
length	141.10	155.90
width	60.30	63.60
height	53.20	52.00
curb_weight	1488	1874
engine_type	l	ohc
num_of_cylinders	three	four
engine_size	61	90
fuel_system	2bbl	2bbl
bore	2.91	3.03
stroke	3.03	3.11
compression_ratio	9.50	9.60
horsepower	48	70
peak_rpm	5100	5400
city_mpg	47	38
highway_mpg	53	43
price	5151	6295
density	5.680554	6.280681

What/which geometric features can affect the mpg of the cars and in what ways?

- Plots and visualization



How can engine properties affect mpg?- Conclusions

- Higher horsepower shows a decrease in both types of MPG.
- Larger engines show a decrease in both types of MPG.
- Number of cylinders shows that 3 and 4 are ideal for best MPG
- Gas vehicles have both the highest and lowest MPG
- Stroke and RPM do not affect MPG

How does each car price and make affect the insurance rating and losses on each vehicle? - Conclusions

- Cars such as Audi, Saab, and Porche all have a relatively high risk rating.
- However, Jaguar, BMW, and Mercedes-Benz, all have lower risk ratings
- Insurance Losses are somewhat tied to price, but not a coherent trend
- The price of the vehicle *may* be tied to price, however the data set did not contain a lot of high-price or extremely high/low risk vehicles
- Conclusion: Insurance will change depending on the make, but not obviously tied to any price trends
- Driver rating will probably play a larger factor than initial insurance risk evaluation

What/which geometric features can affect the mpg of the cars and in what ways? - Conclusion

- Cars with a lower density generally have higher MPG
- Cars with lower density are cheaper and cars with higher density are more expensive
- The Hatchback car body type gets the best MPG
- Of all the car makes, on average, Chevrolet gets the best MPG

Task 7- Reflection (a)

This process on analyzing car data was very exciting because it followed a very real problem someone might have – deciding which car is best for them. By using techniques which now seem very simple to us, we could easily make new insights from the data to form decisions and create more questions. We were surprised that certain engine properties did not correlate do higher or lower MPG, especially stroke and number of cylinders. If we were to do this again, we would have found a dataset that includes more cars, newer cars, and includes data over multiple years. This would allow us to make insights into not only the state of one or a few cars made by a company, but of how the products of a company evolve over time.

Task 7- Reflection (b)

We all collectively have a much better perspective on how to analyze data effectively and within a reasonable amount of time. As far as study/work habits, they have mostly just stayed constant, however we have all been a bit more proactive with our assignments compared to usual. They have slightly improved. We would potentially improve these by having slightly tighter and more concrete deadlines, however the assignments were incredibly helpful for learning outside of class time.